



CAP 4620: INTEGRATING HPC, AI & SCIENCE GATEWAYS INTO ARTIFICIAL INTELLIGENCE

ABSTRACT

Paste the same up-to-500-word abstract submitted to the conference portal — what you built, the impact on your user community, and lessons learned. No references.

Through FacultyHack@Gateways 2026, I redesigned CAP 4620: Artificial Intelligence at Florida A&M University to bring shared research cyberinfrastructure into an undergraduate AI course that had previously relied on local machines and Google Colab. The redesign integrates the National Research Platform (NRP) as the course's primary computing environment: students launch GPU-accelerated Jupyter notebooks through the NRP JupyterHub, train neural networks and CNNs on CPU versus GPU resources, run hyperparameter experiments in parallel, and compare NRP against Colab to see for themselves when research computing pays off. The final project was rebuilt as an AI + HPC experience in which students execute experiments on NRP, analyze both model and computational performance, and publish a reproducible workflow on GitHub.

This mattered for our students and program because most FAMU computing undergraduates had never touched research cyberinfrastructure before this course. The JupyterHub gateway let them work in a notebook environment they already knew while gaining authentic experience selecting resources, monitoring jobs, and reasoning about performance — skills that connect directly to graduate research and industry AI work, and that strengthen the pipeline our AI & Data Science Cluster is building.

Two honest lessons emerged. First, less is more: focusing on a single environment (NRP) rather than surveying several platforms gave students a deeper, more confident experience — the temptation to showcase everything is real, and resisting it made the course better. Second, cyberinfrastructure onboarding takes longer than expected; account setup, resource selection, and environment configuration deserve dedicated lab time early in the semester, not a quick walkthrough squeezed into an existing lesson.

INTRODUCTION

Course context and what motivated this integration.

CAP 4620 is FAMU's upper-level AI course, a key elective in the department's AI and data science pathway. Before this redesign all ML work ran on laptops or Google Colab, and most students had never touched research cyberinfrastructure — skills expected in graduate research and industry AI. This project integrates the National Research Platform (NRP): GPU-accelerated Jupyter notebooks on shared cyberinfrastructure

Course Goals

- Master core AI/ML concepts — search, neural networks, deep learning, model evaluation — in Python
- Analyze how batch size, architecture, and CPU vs. GPU choices affect training time and performance
- Practice reproducibility: Git, documented environments, published workflows
- Execute GPU-accelerated AI/ML workflows on the NRP JupyterHub

Grading Scale

A	90% – 100%
B	80% – 89%
C	70% – 79%
D	60% – 69%
F	59% and Below

Mentor Feedback Incorporated

- Embed HPC into existing AI/ML topics rather than a standalone unit — led to CPU vs. GPU, CNN training, and parallel hyperparameter labs built into current course topics
- Give students a science gateway experience — led to adopting the NRP JupyterHub as the interface for selecting resources and executing workflows
- Make cyberinfrastructure assessable — led to explicit HPC/gateway learning outcomes, computational performance analysis, and an AI + HPC final project

TARGET COURSE DESCRIPTION

Catalog-style facts: name, number, department, enrollment, prerequisites, repo.

Course Name	Artificial Intelligence
Course Number	CAP 4620 (Section 301)
Department	Computer and Information Sciences
Enrollment	[Anticipated Enrollment]
Prerequisites	[Prerequisites, or None]
Project Repo	github.com/Yohn-Parra

Course Description

An upper-level introduction to AI — intelligent agents, search, knowledge representation, machine learning, and neural networks — with hands-on model building in Python (Russell & Norvig, 4th ed.) on NRP's GPU-accelerated cyberinfrastructure.

Key Content by Week

Weeks 1–2: ML intro, Jupyter, EDA. Weeks 3–6: regression, classification, ensembles; hyperparameter tuning (grid search on NRP). Weeks 7–9: clustering, PCA, anomaly detection; proposals after Week 8. Weeks 10–13: neural nets, CNNs, RNNs, NLP — GPU labs on NRP; NRP vs. Colab. Weeks 14–15: RL, ethical AI, final presentations.

SCIENCE GATEWAY RESOURCES USED

The specific platform(s) used, who operates them, and why they fit.

National Research Platform (NRP) — an NSF-supported national computing platform providing shared CPU/GPU resources via Kubernetes, operated by UC San Diego and partners. Students use the NRP JupyterHub to select resources and launch GPU-accelerated notebooks without touching the underlying infrastructure. Supporting resources: Cornell Virtual Workshop and HPC Carpentry.

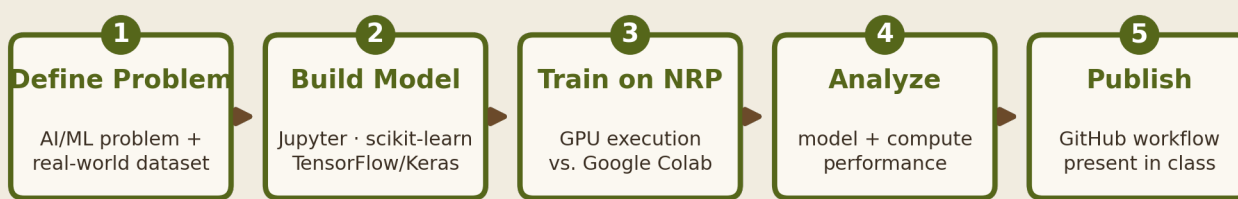


FINAL PROJECT / DELIVERABLE

What students build with the technology, step by step.

AI + HPC Final Project: Reproducible Machine Learning on NRP

- Define an AI/ML problem and prepare a real-world dataset
- Build the model in Jupyter with scikit-learn and TensorFlow/Keras
- Train on NRP GPUs and compare execution against Google Colab
- Analyze model performance and computational performance (time, utilization, memory)
- Publish a reproducible GitHub workflow and present results in class



Project Goals

- Use NRP GPU resources to execute and evaluate computationally intensive AI/ML workflows
- Use a science gateway (NRP JupyterHub) to access resources, execute workflows, and analyze results
- Develop reproducible AI/ML workflows using Git and documented environments

EXPANSION IDEAS

How this integration could grow beyond this offering.

- Extend labs to larger datasets and generative AI / LLM fine-tuning workloads on NRP GPUs
- Feed into the FAMU AI & Data Science Cluster pathway — REU projects and a Generative AI Essentials course
- Extend the model to partner institutions via the AUC Data Science Initiative

REFERENCES

Citations for any textbook, article, or resource used.

S. Russell and P. Norvig, Artificial Intelligence: A Modern Approach, 4th Edition, Pearson, 2021. · National Research Platform, nationalresearchplatform.org.

IMPACT ON STUDENTS

Who benefited, and what changed for them.

Users are undergraduate computing majors in CAP 4620 at FAMU, most of whom had never used research cyberinfrastructure before this course. They gained GPU computing experience, confidence selecting and monitoring computational resources, and a published, reproducible AI workflow on GitHub — access and skills that connect directly to graduate research and industry AI roles and strengthen the department's AI and data science pipeline.

COURSE REDESIGN: BEFORE & AFTER

A side-by-side look at the syllabus before and after this redesign.

INITIAL SYLLABUS

- Covered AI foundations and ML entirely on personal laptops and Google Colab
- Final project asked students to develop an AI application, with no computational-resource requirements
- Missing: access to research cyberinfrastructure, GPU computing, and performance analysis

MODIFIED SYLLABUS

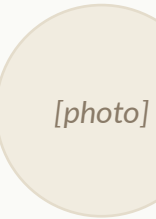
- Added NRP JupyterHub labs: CPU vs. GPU comparisons, GPU-accelerated CNN training, parallel hyperparameter experiments
- NRP is introduced mid-semester and anchors the final AI + HPC project after Week 8
- Assessment now includes computational performance analysis and a published, reproducible GitHub workflow

AUTHORS & MENTORS

Credit the faculty participant and Gateways mentor(s): photo, role, institution, email.



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PROJECT GITHUB REPO

github.com/Yohn-Parra

Sponsors: SGX3 · National Science Foundation · ORNL · Omnibond · TACC · HackHPC



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